

Divy Patel

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Education

Arizona State University (ASU) , Tempe, AZ Master of Science in Computer Science (MSCS) (GPA: 3.55/4)	Aug. 2022 – Dec. 2023
Pandit Deendayal Energy University (PDEU) , India B.Tech in Information and Communication Technology (GPA: 9.22/10)	Apr. 2018 – May 2022

Technical Knowledge

Programming Languages: Java, Python, R, C++, C#, JavaScript, SQL (MySQL, PostgreSQL), GraphQL, HTML
Frameworks: RESTful APIs (Microservices), Django, LangChain, TensorFlow, PyTorch, OpenCV, Scikit-learn
Tools and Technologies: AWS (Lambda, EC2, S3, DynamoDB), Docker, CI/CD, Git, Redis, Jira, Agile, Scrum

Work Experience

GlobalLogic (client: Walmart), Dallas, Texas	Software Engineer	Oct. 2024 – Present
- Developed Python/Django RESTful APIs enabling external TV control through mobile, HDMI-CEC, and IoT devices. - Integrated new hardware abstraction layer into TV OS components, removing manufacturer library dependency across chipsets. - Built a Generative AI-based BDD test generation pipeline for Behave suites, cutting AI cost per module from \$300–400 to \$40–60. - Migrated smartcast-platform-layer TV middleware from Python to MicroPython via AI code transformation, validated by BDD suites. - Automated defect tracking and reporting, saving the team lead over 6 hours weekly. - Consolidated five middleware release branches into one stable branch, eliminating cherry-pick overhead and fix divergence.		
Arizona State University , Tempe, AZ (part-time)	Engineer	Jul. 2024 – Dec. 2024
- Configured Linux servers and installed dependencies for the website, database, and ML needs. - Assessed team requirements and recommended hardware for better performance and cost-efficiency. - Contributed to the development of an LLM to generate daily tasks for artists based on their long-term goals.		
Arizona State University , Glendale, AZ (part-time)	Research Technician	Apr. 2024 – Oct. 2024
- Developed programs to analyze genome data according to the lab's requirements under the supervision of Dr. Sreetharan Kanthaswamy. - Contributed to a project analyzing genetic diversity and mutations across different monkey breeds.		
OleOleSport , Remote (Santa Monica, CA)	Software Engineer	Apr. 2024 – Jul. 2024
- Built a football history chatbot using LangChain, Mistral 7B, and architectures like RAG and CRAG. - Experimented with models, including Llama, achieving 98% accuracy for Arsenal team history Q&A. - Designed and developed a Flutter-based front-end for a football fan community website.		
Quartic.ai , Remote	Data Engineer	Jul. 2021 – Jun. 2022
- Built real-time data pipelines pre-processing live data feeds with Python, GraphQL, and Redis, deployed via CI/CD. - Optimized the data pipeline feeding ML models for predictive analytics and real-time risk monitoring. - Developed a live risk monitoring dashboard on pipeline outputs using Grafana and ReactJS.		
HOPS Healthcare , India	Artificial Intelligence Intern	Apr. 2021 – Dec. 2021
- Achieved 85% accuracy in mobile-based skin disease detection by implementing OpenCV, TensorFlow, and CNNs. - Developed a noise reduction algorithm in Python, aiding detection of heart conditions from stethoscope audio.		

Projects

LLM for Document Based Question Answering 	Project
- Developed Large Language Model (LLM) using RAG implementation to interpret PDFs, CSV, and JSON files, with 98% accuracy. - Used LangChain for efficient storage and retrieval, ensuring accurate responses by implementing Prompt Engineering. - Enhanced chat automation processes by testing model performance on diverse and complex queries.	
Smart Parking Database Management System 	PDEU
- Led development of a comprehensive database management system for smart parking lot application. - Generated 24,000 dummy data entries to thoroughly test and validate the database system. - Documented SQL queries, query output snapshots, and CSV exports for database testing and analysis.	
Graph Operations and Search	ASU
- Developed a Java application for graph operations with DOT parsing, node and edge management, and visualization using OOP patterns. - Implemented Breadth-First Search (BFS) and Depth-First Search (DFS) with unit tests covering core operations. - Provided a command-line interface for graph interaction, searches, and export to .dot and image formats.	

Recent Achievements

Publication in Materials Today: Proceedings on Artificial intelligence powered material search engine	Apr. 2022
Publication in American Journal of Primatology on Sequencing Reveals MHC Diversity in Captive Rhesus Macaque Colonies	Jan. 2026